

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402963
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Daley; Devon Manuel et al.

Apparatus, systems, and methods to facilitate instrument visualization

Abstract

Disclosed embodiments pertain to apparatus, systems, and method for facilitating instrument visualization and use in robotic medical devices. In some embodiments, for one or more instruments coupled to a robotic medical device, one or more corresponding instrument positions relative to a current field of view (FOV) of least one image sensor coupled to the robotic medical device may be obtained. Feedback, which may haptic and/or visual, may be initiated for a selected instrument based on a position of the selected instrument relative to the current FOV.

Inventors: Daley; Devon Manuel (San Mateo, CA), Chen; Zihan (Union City, CA), Romo; Enrique (Danville, CA)

Applicant: Auris Health, Inc. (Redwood City, CA)

Family ID: 74258927

Assignee: Auris Health, Inc. (Santa Clara, CA)

Appl. No.: 16/936419

Filed: July 22, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210030497 A1	Feb. 04, 2021

Related U.S. Application Data

us-provisional-application US 62881346 20190731

Publication Classification

Int. Cl.: A61B34/35 (20160101); A61B18/00 (20060101); A61B34/00 (20160101); G02B6/02 (20060101)

U.S. Cl.:

CPC A61B34/35 (20160201); A61B34/76 (20160201); G02B6/02076 (20130101); A61B2018/00297 (20130101); A61B2562/028 (20130101)

Field of Classification Search

CPC: A61B (34/76); A61B (2018/00297)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8483816	12/2012	Payton	700/245	A61B 34/30
9403056	12/2015	Weinberg	N/A	A63B 21/4045
9582096	12/2016	Narita	N/A	G06F 3/0412
10149720	12/2017	Romo	N/A	A61B 17/00234
10219874	12/2018	Yu	N/A	A61M 25/0012
10730187	12/2019	Larkin	N/A	A61B 1/0051
2005/0273254	12/2004	Malchi	702/152	G01C 21/005
2008/0004603	12/2007	Larkin	606/1	A61B 34/25
2008/0287805	12/2007	Li	600/471	A61B 34/20
2009/0112469	12/2008	Lapidot	701/469	G01C 23/00
2009/0120696	12/2008	Hayakawa	178/18.05	G06F 3/045
2010/0331856	12/2009	Carlson et al.	N/A	N/A
2012/0113030	12/2011	Park	345/173	H04M 1/72469
2013/0211588	12/2012	Diolaiti	700/249	B25J 9/1689
2013/0222339	12/2012	Koga	345/174	G06F 3/016
2014/0276938	12/2013	Hsu	606/130	A61B 34/76
2015/0230697	12/2014	Phee et al.	N/A	N/A
2016/0049217	12/2015	Tee	73/862.627	H01B 1/22
2017/0165013	12/2016	Itkowitz	N/A	A61B 34/35
2017/0202624	12/2016	Atarot	N/A	G16H 40/63
2017/0215823	12/2016	Ivanov	N/A	A61B 6/545
2018/0228343	12/2017	Seeber	N/A	A61B 34/20
2018/0242941	12/2017	Grass	N/A	G06T 7/251
2020/0304753	12/2019	Venkataraman	N/A	A61B 1/00045

Primary Examiner: Jacob; Oommen

Attorney, Agent or Firm: Frost Brown Todd LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of and priority to U.S. Provisional Patent Application No. 62/881,346 entitled, "APPARATUS, SYSTEMS, AND METHODS TO FACILITATE INSTRUMENT VISUALIZATION," filed Jul. 31, 2019, which is assigned to the assignee hereof, and incorporated by reference in its entirety, herein.

FIELD

(1) The subject matter disclosed herein relates to robotic medical systems, devices, and methods to facilitate instrument visualization.

BACKGROUND

(2) Robotic medical systems are often used during minimally invasive or non-invasive medical procedures for imaging tissue, performing biopsies, surgery, and/or other medical procedures. In many instances, these medical procedures can be hampered because instruments associated with the robotic medical system may be outside the Field of View (FOV) of a camera used by the robotic medical system at some stage of the medical procedure. Because cameras on many robotic medical devices used for minimally invasive or non-invasive procedures typically have a limited FOV and/or limited maneuverability, an operator (e.g. a medical professional) may encounter situations where the instruments are out of the FOV relatively frequently. Moreover, the operator may find it difficult to determine a direction of movement for the instrument(s) and/or a movement distance in order to bring the instrument(s) into the FOV of the camera. The inability to see instruments and/or difficulty in determining how to bring instruments into the FOV of the camera can encumber professionals, impact safety, and lengthen procedure time thereby decreasing efficiency and increasing cost. In addition, keeping track of instruments and/or maintaining FOV can significantly increase operator cognitive load and detract focus from substantive procedures. Accordingly, some embodiments disclosed herein enhance safety and improve procedural efficiency, in part by facilitating instrument visualization and control during medical procedures.

SUMMARY

(3) In some embodiments, a method on a robotic medical system may comprise: obtaining, for one or more instruments coupled to a robotic medical device, one or more corresponding instrument positions relative to a current field of view (FOV) of least one image sensor coupled to the robotic medical device; and initiating, for a selected instrument of the one or more instruments, generation of a corresponding feedback to a user, wherein the corresponding feedback is based on a position of the selected instrument relative to the current FOV. In some embodiments, to obtain the one or more corresponding instrument positions relative to the current FOV, the method may comprise: determining, based on a corresponding current pose associated with at least one image sensor coupled to a robotic medical device, the current FOV of the at least one image sensor; and determining the one or more corresponding instrument positions relative to the current FOV based on: one or more images captured by the at least one image sensor, or a control model for the robotic medical device, or sensor input from one or more sensors coupled to the robotic medical device, or a combination thereof.

(4) In another aspect, a robotic medical device may comprise: at least one image sensor; one or more instruments; and a processor communicatively coupled to the at least one image sensor and the one or more instruments, wherein the processor is configured to: obtain, for the one or more instruments, one or more corresponding instrument positions relative to a current field of view (FOV) of the least one image sensor; and initiate, for a selected instrument of the one or more instruments, generation of a corresponding feedback to a user, wherein the corresponding feedback is based on a position of the selected instrument relative to the current FOV. In some embodiments, the robotic medical device may further comprise one or more sensors coupled to the processor, wherein to obtain the one or more corresponding instrument positions relative to the current field of view (FOV), the processor may be configured to: determine, based on a corresponding current pose

associated with the at least one image sensor, the current FOV of the at least one image sensor; and determine, the one or more corresponding instrument positions relative to the current FOV based on: one or more images captured by the at least one image sensor, or a control model for the robotic medical device, or sensor input from the one or more sensors, or a combination thereof.

(5) In a further aspect, a non-transitory computer-readable medium may comprise instructions to configure a processor on a robotic medical device to: obtain, for one or more instruments coupled to a robotic medical device, one or more corresponding instrument positions relative to a current field of view (FOV) of least one image sensor coupled to the robotic medical device; and initiate, for a selected instrument of the one or more instruments, generation of a corresponding feedback to a user, wherein the corresponding feedback is based on a position of the selected instrument relative to the current FOV. In some embodiments, to obtain the one or more corresponding instrument positions relative to the current field of view (FOV), the instructions may configure the processor to: determine, based on a corresponding current pose associated with at least one image sensor coupled to the robotic medical device, the current FOV of the at least one image sensor; and determine, the one or more corresponding instrument positions relative to the current FOV based on: one or more images captured by the at least one image sensor, or a control model for the robotic medical device, or sensor input from one or more sensors coupled to the robotic medical device, or a combination thereof.

(6) In some embodiments, an apparatus may comprise: means for obtaining, for one or more instruments coupled to a robotic medical device, one or more corresponding instrument positions relative to a current field of view (FOV) of least one image sensing means coupled to the robotic medical device; and means for initiating, for a selected instrument of the one or more instruments, generation of a corresponding feedback to a user, wherein the corresponding feedback is based on a position of the selected instrument relative to the current FOV. In some embodiments, means for obtaining, the one or more corresponding instrument positions relative to the current FOV may comprise: means for determining, based on a corresponding current pose associated with the image sensing means, the current FOV of the image sensing means; and means for determining the one or more corresponding instrument positions relative to the current FOV based on: one or more images captured by the image sensing means, or a control model for the apparatus, or sensor means input from one or more sensor means, or a combination thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the invention will be described, by way of example only, with reference to the drawings.

(2) FIG. 1A shows an example diagram illustrating some features of a robotic medical system in accordance with certain embodiments disclosed herein.

(3) FIG. 1B shows a schematic block diagram illustrating functional blocks of a robotic medical system in accordance with certain embodiments disclosed herein.

(4) FIG. 2A shows an example robotic medical device with a flexible articulating main sheath coupled to image sensors and a plurality of instruments in accordance with certain embodiments disclosed herein.

(5) FIG. 2B shows a diagram illustrating image sensors and instruments of an example robotic medical device in a retracted state.

(6) FIG. 2C shows the image sensor and instruments of the example robotic medical device of FIG. 2B in a deployed state.

(7) FIG. 3A shows an example FOV associated with an image sensor coupled to a robotic medical device illustrating potential locations of instruments relative to the current FOV.

(8) FIGS. 3B-3D show various potential locations of instruments relative to the FOV of the image sensor illustrating example visual feedback provided to a user based on the locations of the instruments relative to the current FOV.

(9) FIGS. 4A and 4B show a flowchart of example method to provide user feedback based on the locations of the instruments relative to the current FOV of an image sensor coupled to a robotic medical device.

(10) FIG. 5 shows an exemplary computing subsystem to facilitate instrument visualization.

(11) Like reference numbers and symbols in the various figures indicate like elements, in accordance with certain example embodiments. In addition, multiple instances of an element may be indicated by following a first number for the element with a letter or with a hyphen and a second number. For example, multiple instances of an element **120** may be indicated as **120-1**, **120-2**, **120-3** etc. When referring to such an element using only the first number, any instance of the element is to be understood (e.g. element **120** in the previous example would refer to elements **120-1**, **120-2**, and/or **120-3**).

DETAILED DESCRIPTION

(12) Some disclosed embodiments pertain to robotically driven medical devices and systems and facilitate visualization of instruments during medical procedures such as endoscopy, laparoscopy, and/or endolumenal procedures. In some embodiments, a robotically driven device may comprise a user-controlled robotically driven flexible articulating main sheath (which may also be referred to as a “mother sheath”). One or more robotically driven image sensors and one or more robotically driven instruments may be coupled to a distal end of the main sheath. For example, in some embodiments, the one or more image sensors may be coupled to the distal end of a flexible articulating image sensor sub-arm and a proximal end of the image sensor sub-arm may be coupled to the distal end of the main-sheath. Further, in some embodiments, each instrument may be coupled to the distal end of a corresponding flexible articulating instrument sub-arm and a proximal end of the corresponding instrument sub-arm may be coupled to the distal end of the main-sheath. Thus, in some embodiments, the distal end of the main sheath may be coupled to the proximal end of a plurality of sub-arms, and the distal end of each sub-arm may be coupled to at least one corresponding instrument and/or at least one image sensor. Each flexible articulating (image sensor and/or instrument) sub-arm may be retractable into the main sheath and housed in the main sheath when in a retracted state. The term “sub-arm,” when used without a qualifier, may refer to any type of sub-arm (e.g. either image sensor sub-arm or instrument sub-arms).

(13) In some embodiments, the main sheath and each sub-arm may be robot driven and operation and motion of the main sheath and each sub-arm may be individually controlled by an operator (e.g. a medical professional) through a user-interface. The selection (e.g. of the main sheath and/or one or more image sensors/image sensor-sub arms, and/or one or more instruments/instrument sub-arms), activation (e.g. turning on the image sensor(s)), deployment (e.g. extension and/or retraction of the main sheath and/or each sub-arm), motion (e.g. of the main sheath, each sub-arm, image sensor(s), and/or each instrument), and operation of the image sensor(s) and operation of each instrument may also be individually controlled through the user interface. User input (e.g. using the user interface) may be used to send commands and data to actuators that control the deployment, articulation, motion, and operation of the main sheath, individual sub-arms, the one or more image sensors, and instruments.

(14) In some embodiments, a robotically driven medical system may include endolumenal devices, endoscopic devices, and/or laparoscopic devices. In some embodiments, the robotic medical device may be a single-port device. In “single port” devices, a single entry point on a patient's body may be used to perform medical procedures. The single entry point may be naturally occurring (e.g. oral, anal, etc.) or created via an incision made on the patient's body. For example, the main sheath (e.g. housing retracted sub-arms, image sensors, and instruments) may be inserted through the single port and robotically driven to a location of interest in the patient's body. The image sensors

may be activated when in a housed/retracted state and captured images may be displayed to the user (as the main sheath is being robotically driven to the location of interest) using the visual interface to facilitate navigation of the robotic medical device through the patient's body. Flexibility and articulability of the main sheath may facilitate navigation through the body (e.g. through the gastro-intestinal (GI) tract, respiratory tract, etc.).

(15) Images captured by the image sensor may be displayed to the user (e.g. a medical professional) using a visual interface such as a display or headset. In some embodiments, the display may be a 3-Dimensional (3D) (or stereoscopic) display and the image sensor may provide stereoscopic images. The user interface may also include a haptic interface for an operator of the robotically driven medical device to deploy, control, operate, and move the main sheath, sub-arms, the image sensor(s), and the individual instruments. The haptic interface and/or display may also be used provide feedback to the user. The feedback (haptic and/or visual) may be indicative of device, camera, and/or instrument state as described further herein. Haptic technology may be used to provide the haptic feedback. As used herein, the term haptic feedback refers to any technique of creating a tactile experience, which may include applying forces (e.g. resistance or assistance), vibrations, or other motion to an interface and/or component being operated by a user. Tactile sensors may be used to measure forces exerted by the user on the interface. Electrical, electronic, electromagnetic, and electromechanical techniques may be used to adjust the haptic feedback (e.g. forces, vibration, or motion) based on input from the tactile sensors.

(16) In some embodiments, the user interface may be coupled to the robotically driven medical device (including the main sheath, sub-arms, the image sensor(s), and instruments) through a device control system. The device control system may receive commands from the user interface and control the medical device using a combination of electrical, electronic, electromagnetic, and/or electro-mechanical components.

(17) In some embodiments, for one or more instruments, one or more corresponding instrument positions relative to a current field of view (FOV) of at least one image sensor coupled to the robotic medical device may be obtained; and, for a selected instrument of the one or more instruments, generation of a corresponding feedback to a user may be initiated, wherein the corresponding feedback is based on a position of the selected instrument relative to the current FOV.

(18) In some embodiments, the current FOV for at least one image sensor coupled to the robotic medical system may be obtained based on a corresponding current pose associated with the at least one image sensor. The field of view (FOV) of an image sensor at a point in time refers to the extent of the environment around the image sensor that can be observed by the image sensor. The FOV may be expressed as a solid angle. The image sensor may be sensitive to light (electromagnetic radiation) in the solid angle. FOV, as used herein, may also refer to an angle of coverage, which describes the angular range that the image sensor can effectively image. The angle of coverage for an image sensor may be smaller than the image sensor's theoretical FOV. Image sensor parameters (e.g. focal length, image sensor dimensions, etc.) may be used to determine the angle of coverage.

(19) Further, the corresponding positions of one or more instruments relative to the current FOV may be determined based on: images captured by the at least one image sensor. For example, the corresponding positions of an instrument relative to the current FOV may be determined based on images captured by the at least one image sensor, when the position of the instrument lies within the current FOV. In some embodiments, a control model may be used to determine the corresponding positions of instruments relative to the current FOV based on user motion input from the start of a procedure, and/or from a time of image sensor, instrument, or sub-arm deployment. In some embodiments, sensor input from one or more (optional) sensors coupled to the robotic medical device may be used to determine the corresponding positions of instruments relative to the current FOV (e.g. when one or more of the instruments are outside the FOV).

(20) In some embodiments, feedback may then be provided to the user, wherein the feedback is based on the corresponding positions of instruments relative to the current FOV. The feedback may

comprise haptic and/or visual feedback. For example, the haptic feedback may be provided as haptic resistance to movement of the instrument in response to a determination that the position of the instrument is: (a) within a boundary of the current FOV and (b) within a threshold of the boundary of the current FOV, when the instrument is moved toward the boundary of the FOV. In some instances, the haptic resistance to movement of the instrument toward the boundary of the FOV may be inversely proportional to the distance between the boundary and the position of the instrument when the position of the instrument is within the current FOV and within the threshold of the boundary of the FOV. Haptic resistance may increase the effort or force needed to operate user controls that effect movement of a selected instrument.

(21) As another example, the haptic feedback may be provided as haptic resistance to movement of the instrument in response to a determination that the position of the instrument is: outside the boundary of the current FOV and the instrument is moved further away from the boundary of the current FOV. In some instances, the haptic resistance to movement of the instrument further away from the current FOV boundary may be proportional to the distance between the boundary and the position of the instrument when the position of the instrument is outside the current FOV boundary.

(22) In some embodiments, visual feedback may be provided by displaying a directional indicator indicating a corresponding movement direction for an instrument to bring the instrument further into the current FOV in response to a determination that the corresponding position of the instrument is within the current FOV and within a threshold of a boundary of the current FOV. As another example, visual feedback may be provided by displaying a directional indicator indicating a corresponding movement direction for an instrument to bring the instrument further into the current FOV in response to a determination that the corresponding position of the instrument is outside the boundary of the current FOV. In some embodiments, the magnitude of the directional indicator may be proportional to a distance between the corresponding position of the instrument and a specified location within the current FOV (e.g. the center or estimated center of the current FOV).

(23) In some instances, the robotic medical system may disable functionality associated with an instrument when that instrument is outside the current FOV. For example, some instruments may be deemed or configured as preferably operated with user supervision and functionality associated with these instruments may be disabled when outside the current FOV. In some embodiments, the robotic medical system may be configured to automatically bring an instrument into the current FOV upon user request and/or for the duration of a procedure. In some embodiments, the robotic medical system may be configured to automatically track an instrument so that the instrument stays in the FOV of the image sensor for the duration of a procedure. For example, the image sensor may track instrument motion so that the instrument stays within some threshold of the FOV boundary. Thus, in some embodiments, when appropriately configured, motion of an image sensor may exhibit some correlation to the motion of a corresponding tracked instrument.

(24) Conventional laparoscopic systems, which can be manual or non-robotic, may demand operator dexterity to operate the camera and instruments, which can be challenging. Additionally, conventional systems are limited by physician ergonomics, which can limit the physician's range of motion. For example, in conventional single port systems, many distinct instruments may be inserted through a single port, limiting the available range of motion for the camera and/or instruments thereby increasing procedural difficulty. Multi-port systems, which require multiple incisions (e.g. one incision for each instrument), may allow increased range of motion, but increase recovery time. Moreover, in multi-port systems, ensuring that the distinct instruments remain in the camera FOV or bringing instruments into the FOV can be difficult and significantly increase operator cognitive load. In addition, in some conventional systems, an instrument may be coupled to a main arm that may be rigid and/or not independently maneuverable. The rigidity and/or non-maneuverability of the main arm may limit or increase the difficulty of many types of procedures that may be performed with conventional systems. For example, the GI tract includes many curves

and bends, so a rigid and stiff main arm may limit procedures to the small straight section of the colon, the rectum, significantly reducing the capability of the device, and/or increasing the risk of tissue rupture.

(25) FIG. 1A shows an example diagram illustrating some features of a robotic medical system **100** in accordance with certain embodiments disclosed herein.

(26) In some embodiments, robotic medical system **100** may comprise user interface **130** (which may take the form of a user console), which may be electrically, electro-mechanically, and/or communicatively coupled to robotic medical device **200**. Embodiments of robotic medical device **200** are described further in relation to FIGS. 2A-2C below. As shown in FIG. 1A, robotic medical device **200** may comprise main sheath **235**, image sensors **110**, and instruments **120** (e.g. instruments **120-1** and **120-2**).

(27) In some embodiments, the communicative coupling between user interface **130** and robotic medical device **200** may occur over a communications interface, which may be wired (e.g. wired communications interface **107**) or wireless (e.g. wireless communication interface **105**). The term “communications interface,” as used herein, may refer to a wired communications interface or a wireless communications interface. For example, commands input by user using user interface **130** may be wirelessly communicated (e.g. over wireless communication interface **105**) to a robotic medical device control system (e.g. robotic medical device control system **160** shown in FIG. 1B) associated with robotic medical device **200**. In some embodiments, the robotic medical device control system may control and drive robotic medical device **200** based on commands received over the communications interface. Wired communication may occur using wired networks (including over the Internet and/or private networks), fiber optic interfaces, other widely available interfaces such as Universal Serial Bus (USB), Ethernet, Thunderbolt, and/or other proprietary interfaces. In some embodiments, wireless communication interfaces may include Wireless Personal Area Networks (WPANs) (e.g. based on the IEEE 802.15x standards) which may facilitate wireless communication between devices over short distances (e.g. within a room). Wireless communication may also include communication over Wireless Local Area Networks (WLAN), which may be based on the IEEE 802.11 standards, and/or over Wireless Wide Area Networks (WWAN), which may be based on cellular communication standards such as a Fifth Generation (5G) network, or Long Term Evolution (LTE). 5G and LTE based communication are described in documents available from an organization known as the 3rd Generation Partnership Project (3GPP). In some embodiments, a combination of wired and wireless communications may be used.

(28) Thus, in some embodiments, a user interface **130** may be remotely situated from robotic medical device **200**, and robotic medical device **200** may be controlled and operated based on input received by robotic medical device control system from user **130** over the communications interface. The robotic medical device control system (not shown in FIG. 1A) may control actuators and/or other electronic, electrical, electromagnetic, and/or electro-mechanical components associated with robotic medical device **200** based on the received commands (e.g. from user interface **130** over communications interface). In some embodiments, robotic medical system **100** may also include electrical coupling between user interface **130** (e.g. a user console) and robotic medical device **200** so that user interface **130** (e.g. the user console) may be electrically and communicatively coupled to robotic medical device **200** using some combination of wired and/or wireless links.

(29) User interface **130** may comprise example haptic interface **140-1** and **140-2** (collectively referred to as haptic interface **140**) and visual interface **135**. In some embodiments, visual interface **135** may be stereoscopic and may comprise a 3D display and provide a 3D view of the environment around robotic medical device **200** (e.g. in embodiments where robotic medical device **200** includes stereoscopic image sensors **110**). In some embodiments, In some embodiments, visual interface **135** may alternatively or additionally include a head mounted display. Visual interface **135** may also display an indication of the location of the robotic medical device **200** within the

patient's body at various levels of granularity. Robotic medical device controls **145-1**, **145-2**, **145-3**, and/or **145-4**, (shown as foot pedals in FIG. **1A**) may be used by the user to activate, deploy, select, control, and/or move one or more of main sheath **235**, the sub-arms, image sensors **110**, and/or instruments **120**. For example, a user may use robotic medical device controls **145** (e.g. foot pedals) to select and activate an instrument **120**, prior to moving the instrument **120** via haptic interface **140**. Once the instrument **120** has been moved to a desired location, robotic medical device controls **145** (and/or haptic interface **140**) may also be used to control instrument motion, instrument function, and perform procedures. In some embodiments, the selected component and/or the current function being performed using robotic medical device controls **145** may be displayed or indicated to the user (e.g. as an overlay) in visual interface **135**.

(30) Haptic interfaces **140** may each have degrees of freedom that reflect the degrees of freedom available to mother sheath **235**, and/or the individual sub-arms, and/or image sensors **110**, and/or instruments **120** on robotic medical device **200**. In some embodiments, upon selection of one of the above components of robot medical device **200**, haptic interface **140** may reflect the degrees of freedom available to the selected component. As shown in FIG. **1A**, haptic interfaces **140-1** and **140-2** may be moved and/or oriented in various directions, as indicated in FIG. **1A** by directional arrows reflecting motion related user input **143-1** and **143-2**, respectively. Haptic interfaces **140** may also include other mechanisms to provide user input (e.g. triggers, buttons, thumb wheels, etc.) to control the mother sheath **235**, the various sub-arms individually, the instruments **120** and/or the image sensors **110**. In some embodiments, one haptic interface (e.g. **140-2**) may be used to articulate/move a selected component (e.g. mother sheath **235**, sub-arms, selected instrument **120**, and/or image sensors **110**), while the other (e.g. **140-1**) may be used to exercise control over the function of the instrument **120** and/or to provide other input.

(31) The movements of haptic interfaces **140** (e.g. by a user) may be mirrored by one or more of mother sheath **235**, sub-arms, image sensors **110**, and/or instruments **120** on robotic medical device **200** (depending on the currently active and/or selected component(s)). In some embodiments, users may configure robotic medical system **100** to a desired sensitivity so that movements of selected and/or active instruments appropriately reflect user movement of haptic interfaces **140**. The term “degrees of freedom” refers to the number of independent parameters that determine the pose of an object. The term “pose” refers to the position (e.g. X, Y, Z coordinates) and orientation (e.g. roll, pitch, and yaw) of an object relative to a frame of reference. Pose may be specified as a 6 Degrees-of-Freedom (DOF) pose, which may include positional coordinates (e.g. X, Y, Z) and orientation information (e.g. roll, pitch, and yaw) relative to a frame of reference. In some embodiments, haptic interface **140** may be used to articulate, move, and orient one or more of main sheath **235**, the individual sub-arms, image sensors **110**, and/or instruments **120-1** and **120-2**. For example, haptic interface **140** may be used to place image sensors **110** in a specified pose. The term “camera pose” or “image sensor pose” may refer to the position and orientation of the image sensor relative to a frame of reference. The image sensor pose may be used to determine a field of view of the image sensor. The field of view of the image sensor may be expressed in mathematical form (e.g. as a conical section). In some embodiments, the frame of reference may be image sensor centric and may be used to express: (a) the field of view of the image sensor, relative to the (image sensor centric) frame of reference and/or (b) the position and orientation of one or more instruments **120**. Instrument positions may thus be determined relative to the field of view (e.g. whether inside the FOV, outside the FOV, location relative to a boundary of the FOV, etc.).

(32) In some embodiments, haptic interfaces **140** may also provide haptic feedback **141** (FIG. **1B**) to the user. Tactile sensors may be used to measure forces exerted by the user on the interface. Electrical, electronic, electromagnetic, and electromechanical techniques may be used to adjust the haptic feedback (e.g. forces, vibration, or motion) based on input from the tactile sensors. In some embodiments, the haptic feedback **141** provided to the user may be based on the corresponding positions of instruments **120** relative to the current FOV of image sensors **110**. For example, the

haptic feedback **141** may be provided as haptic resistance to movement of the instrument **120** in response to a determination that the position of the instrument **120** is: (a) within a boundary of the current FOV and (b) within a threshold of the boundary of the current FOV, when the instrument is moved toward the boundary of the current FOV. In some instances, the haptic resistance to movement of the instrument **120** toward the boundary may be inversely proportional to the distance between the boundary and the position of the instrument **120** when the position of the instrument **120** is within the current FOV and within the threshold of the boundary of the current FOV. In some embodiments, as outlined above, haptic interface **140** may include buttons, triggers, or other mechanisms to accept user input. For example, the user may configure robotic medical device **200** (e.g. by setting parameters in configuration information **177**) so that one or more instruments **120** may be automatically positioned (e.g. centered) within the current FOV upon user input—such as by activating a trigger or pressing a button on haptic interface **140**.

(33) As another example, the haptic feedback **141** may be provided as haptic resistance to movement of the instrument **120** in response to a determination that the position of the instrument **120** is: outside the boundary of the current FOV and the instrument **120** is moved further away from the boundary of the current FOV. In some instances, the haptic resistance to movement of the instrument **120** further away from the current FOV boundary may be proportional to the distance between the boundary and the position of the instrument **120** when the position of the instrument **120** is outside the current FOV boundary.

(34) Thus, the haptic feedback **141** may provide the user with a real-time interactive indication of the current FOV boundary, when the user attempts to move one or more instruments **120** in direction that would result in the instruments **120** being out of the current FOV of the image sensors **110** or closer to the edge of the FOV (when the instruments **120** are currently in view), or when one or more instruments **120** not currently in the FOV are moved in a direction further away from the current FOV.

(35) In some embodiments, visual interface **135** may also be used to provide visual feedback **117** (FIG. **1B**). For example, visual feedback **117** may be provided by displaying a directional indicator indicating a corresponding movement direction to bring one or more instruments **120** further into the current FOV in response to a determination that the corresponding position of the instrument **120** is within the current FOV and within a threshold of a boundary of the current FOV. As another example, visual feedback **117** may be provided by displaying a directional indicator (e.g. an arrow) indicating a corresponding movement direction for an instrument **120** to bring the instrument **120** into the current FOV in response to a determination that the corresponding position of the instrument **120** is outside the boundary of the current FOV. In some embodiments, the magnitude of the directional indicator may be proportional to a distance between the corresponding position of the instrument **120** and a specified location within the current FOV (e.g. the center or estimated center of the current FOV).

(36) FIG. **1B** shows an example schematic block diagram illustrating some functional blocks of a robotic medical system in accordance with certain embodiments disclosed herein. As shown in FIG. **1B**, robotic medical system **100** may comprise visual interface **135** (e.g. a display headset, a 3D display or stereoscopic display, etc.), haptic interface **140**, and robotic medical device controls **145**. Visual interface **135**, haptic interface **140**, and robotic medical device controls **145** have been described above in relation to FIG. **1A**).

(37) Robotic medical system **100** may further include processor(s) **150**, memory **170**, robotic medical device control system **160**, and robotic medical device **200** (as described further below in relation to FIGS. **2A-2C**). FIG. **1B** is merely exemplary and the functionality associated with blocks shown in FIG. **1B** may be combined (e.g. into a single block), or the functionality in a block may be distributed across several blocks. For example, the functionality associated with robotic medical device control system block **160** may be integrated within processor(s) **150** block or vice versa. As another example, the functionality associated with processor(s) **150** block and user

interface block **130** may be combined. As a another example, the functionality associated with processor(s) **150** block, user interface block **130**, and robotic medical device control system block **160** may be combined. As a further example, the functionality associated with processor(s) **150** block may be distributed between robotic medical device control system block **160** and user interface block **130**.

(38) Haptic Interface **140** may provide motion-related user input **143** to processor(s) **150** based on user motion. As outlined above, user motion may be received by processor(s) **150** as motion-related user input **143** and communicated to robotic medical device control system **160**, which may process the received input and provide appropriate signals to articulate/move the selected and/or active component (one or more of main sheath **235**, sub-arms, image sensors **110**, and/or instruments **120**). Haptic feedback **141** (e.g. based on the position of instruments **120** relative to the current FOV of image sensors **110**) may be provided to the user via haptic interface **140**.

(39) User operation of robotic medical device controls **145** (e.g. as described above in relation to FIG. **1A**) may provide robotic medical device control input **147** to processor(s) **150**, which may be processed and communicated to robotic medical device control system **160**. Robotic medical device control system **160** may further process the received robotic medical device control input **147** and provide as motion control/instrument control input **162** to robotic medical device **200**. Motion control/instrument control input **162** may include appropriate signals to select, activate, deploy, retract, disable, and/or enable a robotic medical device component (e.g. one or more of main sheath **235**, image sensors **110**, and/or instruments **120**).

(40) In some embodiments, processor(s) **150** may be coupled to memory **170**. As shown in FIG. **1B**, memory **170** may include control model **175**, which may be used by processor(s) **150** to determine the positions of instruments **120** relative to image sensors **110**. In some embodiments, control model **175** may include calibrated instrument control models, which may be used to estimate instrument position relative to an image sensor pose and/or a current FOV of image sensors **110**. In some embodiments, control model **175** may be configured to make use of information in captured images **115** (e.g. when one or more instruments are visible in the FOV of image sensor **110**) when being used (e.g. by processor(s) **150**) to determine instrument positions relative to the FOV of image sensors **110**.

(41) In some embodiments, images captured by image sensors **110** may be processed using object recognition and/or tracking techniques (e.g. by processor(s) **150**) to determine a location of one or more instruments **120** in the current FOV of image sensor **110**. In some embodiments, based on configuration information **177** (e.g. as set or invoked by a user), object tracking techniques may also be used to keep an instrument **120** in the FOV of image sensors **110** during a procedure (or portion of a procedure). Object tracking techniques may use one or more of: control model **175**, the known instrument form factor and/or other instrument characteristics, in conjunction with image processing and computer vision techniques to locate an instrument **120** in captured images **115**. Image sensors **110** may be moved to keep the instrument **120** in the FOV of image sensors **110**. Because the form factors of instruments **120** coupled to robotic medical device **200** are known, object recognition and tracking techniques may be applied to identify instruments **120** in a sequence of image(s) captured by image sensors **110** and determine their respective locations relative to the image sensors and/or the corresponding image sensor FOV. In some embodiments, instruments **120** and/or sub-arms may include known markers (e.g. registration markers) to facilitate instrument identification, tracking, and location determination (e.g. relative to the camera centric frame of reference).

(42) In some embodiments, sensor/actuator information **167**, instrument state **165**, and captured images **115** may be provided (e.g. by robotic medical device control system **160**) to processor(s) **150** and/or user interface **130**.

(43) In some embodiments, instrument positions **119** may be determined (e.g. by processor(s) **150** using control model **175**) based on one or more of: sensor/actuator information **167**, configuration

information **177**, prior motion-related user input **143**, and/or instrument state **165**. In some embodiments, instrument positions **119** (e.g. as determined by processor(s) **150** and/or robotic medical device control system **160**) may be made provided to user interface **130**. In some embodiments, processor(s) **150** and/or user interface **130** may determine instrument positions **119** independently based on one or more of: sensor/actuator information **167**, configuration information **177**, prior motion-related user input **143**, and/or instrument state **165** (e.g. by invoking control model **175**).

(44) Configuration information **177** may provide information pertaining to the instruments **120** on robotic medical device **200**, image sensor configuration (e.g. lens focal length and other parameters), user preferences (e.g. sensitivity to user movement, the desired level of haptic feedback **141**, automatic maintenance of FOV over instruments **120**, display parameters, etc.) and/or an operational configuration or mode of operation of robotic medical system **100**.

Configuration information **177** may further indicate whether functionality (or a portion of the functionality) of one or more instruments **120** is to be disabled when not in the current FOV of image sensor(s) **110**.

(45) In some embodiments, control model **175** may also use sensor/actuator information **167** from one or more sensors and actuators (when present) in robotic medical device **200**. Sensor/actuator information **167** may provide information pertaining to the state of sensors/actuators coupled to robotic medical device **200**. Sensor/actuator information **167** may be used by control model **175** to determine the corresponding current poses (positions and orientations) of instruments **120**. The sensors coupled to robotic medical device **200** may include one or more of: electronic sensors; electromagnetic sensors; electro-mechanical sensors, including micro-electro mechanical sensors (MEMS). The sensors may be used to sense actuator articulation/motion of the main sheath, and/or of image sensors **110** and/or the image sensor sub-arm, and/or instruments **120** and/or the instrument sub-arms. The sensors may include 3D shape sensing fiber optic sensors; fiber optic force and/or pressure sensors such as photonic crystal fiber (PCF) sensors or Fiber Bragg Grating (FBG) sensors, or make use of scattering arising from FBG sensors, inherently present, or make use of post-process produced Rayleigh scattering. In some embodiments, sensor/actuator information **167** from one or more sensors may be used in conjunction with captured images **115** and/or user motion input to determine instrument pose or instrument positions. In some embodiments, the instrument pose and/or instrument positions determined above may be relative to the image sensor centric frame of reference.

(46) In some embodiments, the sensors coupled to robotic medical device **200** may form part of a tracking and pose determination system. Electromagnetic sensors may be embedded in instruments **120** and/or at one or more locations in the image sensor sub-arm and instrument sub-arms.

Electromagnetic sensors may use an electromagnetic field generator and small electromagnetic coils to track the instruments. Input from the electromagnetic sensors may be processed (e.g. using signal processing techniques) to determine and track the poses of one or more instruments **120**. In some embodiments, signal processing techniques may compensate for distortions in sensor readings that may be caused by the presence of non-magnetic conductive materials in the environment. Electromagnetic tracking and pose determination techniques may operate to determine pose even in situations where there is no line of sight to instruments **120** and/or when one or more instruments are outside the FOV of image sensors **110**. In some embodiments, input from the electromagnetic sensors may be used by control model **175** to determine a pose (or relative pose) of one or more instruments **120**.

(47) In some embodiments, captured images **115** (e.g. by image sensors **110**) may be processed using object recognition and/or tracking techniques (e.g. by processor(s) **150**) to determine a location of one or more instruments **120** in the current FOV of image sensors **110**. Because the form factors of instruments **120** coupled to robotic medical device **200** and their corresponding deployment state are known, the above information may be used processor(s) **150** executing control

model **175** to determine locations of instruments **120** coupled to robotic medical device **200** in captured images **115**, and/or to track the positions of the instruments over a sequence of images without the need for external markers. As another example, instruments **120** and/or instrument sub-arms may include registration marks, which may be detected in captured images **115** and used to track instruments and/or determine pose. Thus, tracking and pose determination techniques may be markerless (e.g. based on the known form factors of instruments) or involve the use of registration marks. In some embodiments, control model **175** may be used to determine instrument pose based on some combination of captured images **115**, and/or sensor actuator information **167** (which may include input from one or more of electromagnetic sensors, electro-mechanical sensors, electronic sensors, 3D shape sensing fiber optic sensors; fiber optic force and/or pressure sensors etc.).

(48) In some embodiments, image processing techniques such as object recognition and tracking techniques may be applied to identify instruments **120** in image(s) captured by image sensors **110** and to determine their respective locations relative to the image sensors and/or the corresponding image sensor FOV. Object recognition and tracking techniques may include feature extraction from images, feature matching/tracking, image comparison, image matching, image differencing, pattern recognition, etc. As outlined above, in situations where: (a) there is no line of sight to instruments **120** (e.g. line of sight is obscured by tissue or other obstructions), there is no line of sight to instruments **120**; and/or (b) one or more instruments are outside the FOV of image sensors **110**, then electromagnetic tracking and pose determination techniques may operate to track instruments **120** and determine pose

(49) In some embodiments, processor(s) **150** may use control model **175** and one or more of: motion-related user input **143**, robotic medical device control input **147**, captured images **115**, configuration information **177**, and/or sensor/actuator information **167** to determine a pose of image sensor **110** and a current FOV of image sensor **110** relative to a frame of reference, and/or the position of instruments **120** relative to the current FOV of image sensor **110**.

(50) Although shown as separate from processor(s) **150**, memory **170** may be external and/or internal to processor(s) **150** and may include primary and/or secondary memory. Program code may be stored in memory **170**, and read and executed by processor(s) **150** to perform the techniques disclosed herein. As used herein, the term “memory” refers to any type of long term, short term, volatile, nonvolatile, or other memory and is not to be limited to any particular type of memory or number of memories, or type of media upon which memory is stored. Examples of storage media include computer-readable media encoded with databases, data structures, etc. and computer-readable media encoded with computer programs. Computer-readable media may include physical computer storage media. A storage medium may be any available medium that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise Random Access Memory (RAM) and variations thereof including Non-Volatile RAM (NVRAM), Read Only Memory (ROM) and variations thereof Erasable Programmable (EPROM), Flash Memory, etc. Computer-readable media may also include Compact Disc ROM (CD-ROM), memory cards, portable drives, or other optical disk storage, magnetic disk storage, solid state drives, other storage devices, or any other medium that can be used to store desired program code in the form of instructions and/or data structures and that can be accessed by a computer; disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

(51) In some embodiments, processor(s) **150** may process robotic medical device control input **147** and motion-related user input **143**. The processed information may be sent to robotic medical device control system **160** (e.g. via wireless communications interface **105** and/or wired communications interface **107**). Robotic medical device control system **160** may process the information received from processor(s) **150** and send signals to appropriate actuators on robotic

medical device **200** to control, articulate/move, retract, deploy, and/or invoke functionality associated with one or more of: main sheath **235**, sub-arms, image sensors **110**, and/or instruments **120**.

(52) Although shown in FIG. **1B** as discrete blocks, processor(s) **150** and/or memory **170** may be distributed between robotic medical device control system **160** and user console **130** (e.g. comprising Visual Interface **135**, haptic interface **140**, and robotic medical device control **145**). In one embodiment, robotic medical device control system **160** and user console **130** may each have individual local processors **150**. For example, when user console **130** is remotely situated from robotic medical device **200**, user console **130** and robotic medical device **200** may each have local processors **150**. Accordingly, in one embodiment, local processors associated with user console **130** may be configured to: (a) obtain and transmit motion-related user input **143** and robotic medical device control input **147** to local processors associated with robotic medical device control system **160**; and (b) receive and display captured images **115** and provide visual feedback **117** using visual interface **135**, and provide haptic feedback **141** based on input received from robotic medical device control system **160** (e.g. one or more of: instrument state **165**, sensor/actuator information **167**, captured images **115**, and/or image sensor pose, FOV, and instrument position). Conversely, local processors associated with robotic medical device control system **160** may be configured to: (a) receive motion-related user input **143** and robotic medical device control input **147** from local processors associated with user console **130** and provide appropriate motion control/instrument control input **162** to robotic medical device **200** (e.g. based on the received motion-related user input **143** and robotic medical device control input **147**); and (b) obtain and transmit captured images **115**, instrument state **165**, and sensor/actuator information **167** (and/or determined image sensor pose, current FOV, and instrument position information) to local processors associated with user console **130**. As another example, memory **170** and functionality associated with processor(s) **150** may be shared between user console **130** and robotic medical device control system **160**.

(53) In some embodiments, robotic medical device control system **160** may also obtain sensor/actuator information **167** from sensors/actuators on robotic medical device **200**, captured images **115** from image sensors **110**, and instrument state **165**. Sensor/actuator information **167**, captured images **115**, and instrument state **165** may also be received by processor(s) **150** either directly (when coupled to robotic medical device **200**) or indirectly from robotic medical device control system **160** (e.g. over wireless communication interface **105**/wired communications interface **107**). Robotic medical device control system **160** may control actuators and/or other electronic, electrical, and electro-mechanical components associated with robotic medical device **200** based on the received commands (e.g. motion-related user input **143** and robotic medical device control input **147**). In some embodiments, robotic medical device control system **160** may include functionality to detect and prevent collisions, entanglements, and/or physical contact between sub-arms. For example, sections of the main sheath and/or one or more sub-arms may be reconfigured (e.g. without change to the final poses of the image sensors **110** and/or instruments **120**) to avoid collisions, contact, or entanglement.

(54) The methodologies described herein may be implemented in hardware, firmware, software, or any combination thereof. For a hardware implementation, the processor(s) **150** may be implemented within one or more application specific integrated circuits (ASICs), digital signal processors (DSPs), image processors, digital signal processing devices (DSPDs), programmable logic devices (PLDs), field programmable gate arrays (FPGAs), processors, controllers, micro-controllers, microprocessors, electronic devices, other electronic units designed to perform the functions described herein, or any combination thereof. In some embodiments, processor(s) **150** may include capabilities to: determine a FOV for image sensors **110**, determine a pose of image sensor **110**, determine a FOV associated with image sensors **110**, process images, determine and track corresponding positions of instruments relative to a current image sensor FOV, provide input to control haptic feedback **141**, provide visual feedback **117**, and provide input to control actuators

on robotic medical device **200**. Processor(s) **150** may also include functionality perform other well-known computer vision and image processing functions such as feature extraction from images, image comparison, image matching, object recognition and tracking, image compression and decompression, etc.

(55) FIG. 2A shows image sensors **110** and instruments **120** coupled to the distal end **230** of a flexible articulating main sheath **235** of a robotic medical device **200** in accordance with certain embodiments disclosed herein. In some embodiments, robotic medical device **200** may form part of an endolumenal device, an endoscopic device, or a laparoscopic device. Robotic medical device **200** may form part of a robotically driven medical system or another robotic device and may facilitate visualization of instruments **120** during medical procedures

(56) In some embodiments, a robotic device may include robotic medical device **200**, user-interface **130** (not shown in FIG. 2A) to facilitate user interaction with robotic medical device **200**, and processor(s) **150** (not shown in FIG. 2A) coupled to robotic medical device **200** and the user interface **130**.

(57) Further, as shown in FIG. 2A, robotic medical device **200** may include: (a) flexible articulating main sheath **235**; (b) flexible independently articulable image sensor sub-arm **215**, wherein a proximal end **217** of the image sensor sub-arm **215** is coupled to a distal end **230** of the main sheath **235**; (c) a plurality of flexible independently articulable instrument sub-arms **225** (e.g. **225-1** and **225-2**), wherein each instrument sub-arm (e.g. **225-1** or **225-2**) has a corresponding proximal end **227** (e.g. **227-1** or **227-2**, respectively) coupled to the distal end **230** of the main sheath **235**; (d) at least one image sensor **110** (with image sensors **110-1** and **110-2**), wherein the at least one image sensor **110** is coupled to a distal end **210** of the image sensor sub-arm **215**; and (e) a plurality of instruments **120** (e.g. **120-1** and **120-2**), wherein each instrument (**120-1** or **120-2**) is coupled to a distal end **220** (e.g. **220-1** or **220-2**, respectively) of a corresponding instrument sub-arm (e.g. **225-1** or **225-2**, respectively) of the plurality of instrument sub-arms **225** (e.g. **225-1** and **225-2**).

(58) The processor(s) **150** on the robotic device may be configured to determine, based on a corresponding current pose associated with the at least one image sensor **110**, a current FOV of the at least one image sensor **110**; determine, for an instrument (**120-1** or **120-2**) in the plurality of instruments (e.g. **120-1** and **120-2**), a position of the instrument (**120-1** or **120-2**) relative to the current FOV based, at least in part, on captured images **115** captured by the at least one image sensor **110**; and provide feedback using the user-interface **130**, wherein the feedback is based on the corresponding position of the instrument (**120-1** or **120-2**, respectively) relative to the current FOV. The user interface **130** may include at least one of: a haptic interface **140**, and/or a visual interface **135** and the feedback may include haptic feedback **141** using provided using the haptic interface, **140** or visual feedback **117** using the visual interface **135**; or a combination of visual feedback **117** and haptic feedback **141**. A proximal end of main sheath **235** (not shown in FIG. 2A) may be coupled (electrically, electronically, electromagnetically, and/or electro-mechanically) to robotic medical device control system **160** and/or processor(s) **150**. Embodiments of robotic medical device **200** are described further below.

(59) For example, as shown in FIG. 2A, robotic medical device **200** may comprise a user-controlled robotically driven flexible articulating main sheath **235** (also referred to as “mother sheath”). Main sheath **235** may be selected and controlled by the user using processor(s) **150**. In some instances, processor(s) **150** may form part of user interface **130** and/or robotic medical device control system **160**. In some embodiments, main sheath **235** may be selected and controlled by the user using user interface **130**, which may include robotic medical device controls **145** (FIG. 1A) and/or and haptic interface **140** (e.g. haptic interfaces **140-1** and/or **140-2**).

(60) Main sheath **235** may be flexible, independently articulable at various locations on main sheath **235**, and capable of movement in various directions based on user input provided, for example, by motion-related user input **143** using haptic interface **140**. Main sheath **235** and sub-arms (**215** and/or **225**) may be articulated independently. Further, the movement of a sub-arm (**215**

and/or **225**) may occur independently (e.g. without movement) of main sheath **235** and without movement of another sub-arm. Main sheath **235**, image sensor sub-arm **215**, and instrument sub-arms **225** may each include actuators at various locations to facilitate independent articulation and/or movement of the main sheath and/or corresponding sub-arm. Further, in some embodiments, actuators may facilitate pivoting (e.g. pitch) of image sensor **110** without articulation/motion of image sensor sub-arm **215** and instruments **120** may be capable of pivoting (pitch) and/or rotation (roll) without articulation/motion of the corresponding instrument sub-arms **225**. Actuators may (directly or indirectly) receive individual commands from processor(s) **150** based on user-input (e.g. provided using user-interface **130**).

(61) Main sheath may be recessed and capable of housing sub-arms and image sensors **110**/instruments **120** coupled to the sub-arms when the sub-arms are in a retracted state. Main sheath may also house wires, cables, actuators, sensors etc. For example, image sensor sub-arm **215** and each instrument sub-arm **225** may be retracted and housed when retracted (e.g. in recesses) in main sheath. In some embodiments, input provided to actuators may be used along with control model **175** to determine corresponding positions of instruments **120** relative to a current pose of image sensors **110**. In some embodiments, input from electro-mechanical sensors coupled to the actuators and/or sub-arms may be obtained by processor(s) **150** and used, in part, to determine corresponding positions of instruments relative to a current pose of image sensors **110**.

(62) Thus, main sheath **235** may be articulated (bent in a user-controlled manner at one or more specific locations on the main sheath **235**), moved in various directions, and/or rotated by the user using haptic interface **140** and/or robotic medical device controls **145**. For example, articulating the main sheath **235** at one or more locations may facilitate navigation/positioning of image sensors **110** and instruments **120** in a patient's gastro-intestinal (GI) tract close to an area of attention in the GI tract.

(63) In some embodiments, as shown in FIG. 2A, a distal end **230** of main sheath **235** may be coupled to a proximal end **217** of flexible articulating image sensor sub-arm **215**, and to corresponding proximal ends **227-1** and **227-2** of flexible articulating instrument sub-arms **225-1** and **225-2**, respectively. As shown in FIG. 2A, image sensor sub-arm **215** may be flexible and independently capable of articulating in various directions. A distal end **210** of the image sensor sub-arm **215** may be coupled to one or more image sensors **110**. For example, as shown in FIG. 2A, image sensors **110-1** and **110-2** are coupled to the distal end **210** of image sensor sub-arm **215**. Image sensor sub-arm **215** may be selected and controlled by the user using robotic medical device controls **145** (FIG. 1A) and haptic interface **140** (e.g. haptic interfaces **140-1** and/or **140-2** in FIG. 1A). For example, image sensor sub-arm **215** may be independently (e.g. independently of any other sub-arm **225** and/or independently of main sheath **235**) articulated (bent in a user-controlled manner at one or more specific locations on the image sensor sub-arm **215**), retracted and housed when retracted within main sheath **235**, or operationally deployed. FIG. 2A shows image sensor sub-arm **215** in a deployed state.

(64) In some embodiments, as shown in FIG. 2A, distal end **230** of main sheath may be coupled to a proximal end **227-1** of flexible articulating instrument sub-arm **225-1** and to a proximal end **227-2** of flexible articulating instrument sub-arm **225-2**. As shown in FIG. 2A, instrument sub-arms **225-1** and **225-2** may be flexible and capable of articulating independently in various directions. A distal end **220-1** of instrument sub-arm **225-1** may be coupled to instrument **120-1** (shown as a grasper in FIG. 2A). A distal end **220-2** of instrument sub-arm **225-2** may be coupled to another instrument **120-2** (shown as a grasper in FIG. 2A). Instrument sub-arms **225-1** and **225-2** may be selected and controlled by the user using robotic medical device controls **145** (FIG. 1A) and haptic interface **140** (e.g. haptic interfaces **140-1** and/or **140-2**). For example, instrument sub-arms **225-1** (or **225-2**) may be articulated (bent in a user-controlled manner at one or more specific locations on the image sensor sub-arm **225-1** (or **225-2**) and independently of the image sensor sub-arm **215**, and any other sub-arm, and/or main sheath **235**), retracted, or deployed. Instrument sub-arms **225-1**

and **225-2** may each be moved (independently of main sheath **235**) in various directions and/or rotated by the user using haptic interface **140** and/or robotic medical device controls **145**.

Instrument sub-arms **225-1** and **225-2** may be retracted and housed when retracted within main sheath **235**. FIG. 2A shows instrument sub-arms **225-1** and **225-2** in a deployed state.

(65) In some embodiments, robotic medical system **100**, and/or processor(s) **150**, and/or robotic medical device control system **160** (FIG. 1B) may include functionality to detect and prevent collisions, entanglements, and/or physical contact between any of: image sensors **110**, image sensor sub-arm **215**, instrument **120-1**, instrument sub-arm **225-1**, instrument **120-2**, and/or instrument sub-arm **225-2**. For example, sections of the main sheath and/or one or more sub-arms may be reconfigured (e.g. without change to the final poses of the image sensors **110** and/or instruments **120**) to avoid collisions, contact, or entanglement. In some embodiments, robotic medical device **200** may optionally include electro-magnetic sensors, 3D shape sensing fiber optic sensors, fiber optic force and/or pressure sensors such as photonic crystal fiber (PCF) sensors or Fiber Bragg Grating (FBG) sensors, or make use of scattering arising from FBG sensors or inherently present, or make use of post-process produced Rayleigh scattering (not shown in FIG. 2A). Input from one or more of the above sensors may be used to determine a pose (or relative pose) of instruments **120** and/or instrument sub-arms **225**.

(66) Main sheath **235** and/or image sensor **215** may also include light sources to illuminate an environment around robotic medical device **200**. Image sensors **110** may include cameras, CCD sensors, or CMOS sensors, which may transform an optical image into an electronic or digital image and may send captured images **115** to processor **150**. In some embodiments, image sensors **110** may capture color images, which provide “color information,” while “depth information” may be provided by a depth sensor. The term “color information” as used herein refers to color and/or grayscale information. In general, as used herein, a color image or color information may be viewed as comprising 1 to N channels, where N is some integer dependent on the color space being used to store the image. For example, an RGB image may be viewed as comprising three channels, with one channel each for Red, Blue, and Green information. In some embodiments, depth information may be captured using depth sensors (active or passive). The term “depth sensor” is used to refer to functional units that may be used to obtain depth information independently and/or in conjunction with image sensors **110**.

(67) In some embodiments, image sensors **110-1** and **110-2** may form a (passive) stereoscopic image sensor, and may capture stereoscopic images, which may be used to determine depth information. Accordingly, in some embodiments, captured images **115** (FIG. 1B) may include stereoscopic or 3D images with depth information. For example, pixel coordinates of points common to both image sensors (e.g. **110-1** and **110-2**) in a captured image may be used along with triangulation techniques to obtain per-pixel depth information. In some embodiments, the depth information may be used, in part, to determine the pose of instruments **120** relative to image sensors **110**.

(68) Accordingly, in some embodiments, robotic medical device **200** may include at least one image sensor **110** (e.g. a stereoscopic sensor or 3D image sensor), a plurality of instruments **120** (e.g. grasper **120-2**, cautery knife, etc.), and one or more sensors (e.g. some combination of electromagnetic sensors; or electro-mechanical sensors; or 3D shape sensing fiber optic sensors; or fiber optic force sensors; or fiber optic pressure sensors; or a combination thereof). Further, in some embodiments, the electro-mechanical sensors may include micro electro-mechanical sensors (MEMS); and the fiber optic force sensors and the fiber optic pressure sensors may include one or more of: photonic crystal fiber (PCF) sensors or Fiber Bragg Grating (FBG) sensors. The image sensors **110** may be coupled to a distal end **210** of flexible articulating image sensor sub-arm **215**, and instruments **120** may be coupled to corresponding distal ends **220** of flexible articulating instrument sub-arms **225**. The proximal end **217** of flexible articulating image sensor sub-arm **215** and proximal end of flexible articulating instrument sub-arms may be coupled to a distal end **230** of

flexible articulating main sheath **235**. The image sensor sub-arm **215** and the plurality of instrument sub-arms **225** may be retractable into the main sheath and housed in the main sheath (e.g. during deployment, prior to activation, or upon deactivation). Image sensors **110** may be activated even when the image sensor sub-arm **215** is in a retracted state so that an operator of robotic medical device may be able to capture and view images of an environment around robotic medical device **200** even when the image sensor sub-arm is in a retracted state.

(69) Robotic medical device **200** may further include a plurality of actuators (not shown in FIG. 2A) coupled to a flexible articulating main sheath **235**, image sensors **110**, instruments **120**, flexible articulating image sensor sub-arms **215**, flexible articulating instrument sub-arms **225** to control, extend, retract, articulate, and effect movement of the above components. The sensors and actuators may be disposed at various locations on robotic medical device **200** including at various locations on image sensor sub-arm **215** and instrument sub-arms **225** based on design parameters such as the desired accuracy and precision of position detection, level of articulation and/or movement desired. Each actuator may be responsive to commands from a processor(s) **150** or robotic medical device control system **160**, which may be based on input from user interface **130**. The plurality of actuators may facilitate independent articulation and independent motion of the main sheath, the image sensor sub-arm, and the plurality of instrument sub-arms based on the received commands. Actuators may also be used to provide haptic feedback

(70) FIG. 2B shows a diagram illustrating image sensors **110** and instruments **120**, which are coupled to the distal end **230** of a flexible articulating main sheath **235** of a robotic medical device **200**, in a retracted state and housed in main sheath **235**. Image sensor sub-arm **215** (not visible in FIG. 2B) with image sensors **110** and instrument sub-arms **225-4** and **225-5** (not visible in FIG. 2B) with instruments **120-3** and **120-4**, respectively, may be retracted into recesses in main sheath **235** and housed in main sheath **235** when retracted. For example, when robotic medical device **200** is being deployed and moved a location (e.g. within a patient's GI tract **240**), image sensor sub-arm **215** and instrument sub-arms **225-3** and **225-4** may be retracted so that image sensors **110** and instruments **120** are housed within main sheath **235** thereby facilitating maneuverability of the flexible main sheath **235** (through the GI tract **240**) and faster and easier movement and deployment of robotic medical device **200**. In some embodiments, image sensors **110** may be configured to capture images when in a retracted state (e.g. during deployment of main sheath **235**) so that robotic medical device **200** may be visually navigated through the patient's body (e.g. by a medical practitioner) to a location of interest.

(71) FIG. 2C shows the image sensors **110**, instrument **120-4** (shown as a cautery knife in FIG. 2B), and instrument **120-5** (shown as a grasper in FIG. 2B) each individually coupled to the corresponding distal ends (**210**, **220-3**, and **220-4**, respectively) of corresponding flexible articulating sub-arms **215**, **225-4**, and **225-5**, respectively, in a deployed state in accordance with certain embodiments disclosed herein. For example, once a user has articulated and moved main sheath **235** through GI tract **240** to a location of interest, the user may select and deploy one or more sub-arms such as sub-arms **215**, **225-4**, and **225-5**. For example, image sensors **110** may be moved and oriented to capture images of the location of interest (e.g. a colon polyp) and one or more of instruments **120-3** and/or **120-4** may be moved into positions to perform appropriate medical procedures.

(72) As outlined above, in conventional devices, determining the positions of instruments **120** relative to the current FOV of image sensor **110** may be difficult and impose a cognitive load on the operator thereby detracting attention from the substantive procedures being performed. Further, the rigidity of the main sheath may limit the types of procedures that may be performed and/or increase procedural complications/risk.

(73) In some embodiments, the user may configure robotic medical device **200** so that one or more instruments **120** may be automatically positioned within the current FOV upon user input (e.g. by activating a trigger on haptic interface **140** or some other means). In some embodiments,

configuration information **177** (FIG. 1B) may indicate that some instruments **120** (e.g. cautery knife **120-4**) are to be deactivated and/or disabled when outside the boundary of the current FOV to prevent accidental use and/or increase safety. In some embodiments, haptic interface **140** and visual interface **135** may be used to provide haptic feedback **141** and visual feedback **117**, respectively, based on the position of one or more instruments **120** relative to the current FOV of image sensors **110** as described herein. The haptic feedback **141** and/or visual feedback **117** may facilitate maintenance of instrument position within the current FOV of image sensors **110** and/or returning instruments **120** to the FOV when the instruments **120** are outside the boundary of the current FOV. (74) FIG. 3A shows a FOV **300** associated with image sensors **110** coupled to a robotic medical device **200**. FIG. 3A illustrates possible locations of instruments **120** (not shown in FIG. 3A) relative to the current FOV **300** of image sensors **110**. FOV, as used herein, may also refer to an angle of coverage, which describes the angular range that the image sensor can effectively image. As outlined above, the FOV may be determined based on image sensor parameters (e.g. focal length, image sensor size). For example, FOV **300** may be specified or determined based on solid angle **306**. The FOV associated with a pose may also be mathematically specified (e.g. based on a frame of reference such as an image sensor centric frame of reference).

(75) FIG. 3A is merely an example to illustrate some scenarios referred to herein. As shown in FIG. 3A, an instrument (e.g. instrument **120-5**) may be positioned at location T **305** or at location V **310** in current FOV **300**. Further, Location T **305** may be within a first imaging volume **308** of current FOV **300**. Locations within first imaging volume **308** of current FOV **300** are depicted by dashed lines in FIG. 3A. First imaging volume **308** may correspond to solid angle **306_p**. For example, first imaging volume **308** may correspond to locations where imaging may be optimal and/or instruments **120** of robotic medical device **200** have satisfactory range of motion (e.g. based on the procedure being performed) and/or a lower likelihood of leaving current FOV **300** (e.g. when moved during the procedure). For example, configuration information **177** may include information related to first imaging volume **308** for various procedures and/or information to obtain first imaging volume **308**. In some embodiments, first imaging volume **308** (e.g. which may correspond to locations where imaging may be optimal) may be procedure-specific and/or instrument specific to accommodate a range of instrument motion typical for the procedure and/or for the instrument thereby increasing the likelihood that the instrument remains in the FOV when performing the procedure.

(76) FIG. 3A shows locations, such as point V **310**, which may lie between current FOV **300** and first imaging volume **308**. Thus, point V **310** may lie: (a) within current FOV **300**, and (b) within a threshold of the boundary of current FOV **300** (e.g. outside of first imaging region **308**). In some embodiments, the threshold may be a function of the depth **d 302** from image sensor **110** to a plane (e.g. P1) perpendicular to the axis of current FOV **300** that includes the location (e.g. V **310**), where the depth **d 302** is measured in the direction of the axis of current FOV **300**. For example, as described above, in embodiments with stereoscopic image sensors (or with depth sensors), depth information (e.g. depth **d 302**) may be present and/or determined from captured images **115**. In FIG. 3A, locations between current FOV **300** and first imaging volume **308** (e.g. lying within a threshold of the boundary of current FOV **300**) are indicated by the shaded area.

(77) In some embodiments, when an instrument **120** is located at point T **305**, the user may receive visual feedback **117** (e.g. on visual interface **135**, which may be stereoscopic) indicative of direction TA **309** toward center U **303** (or an estimated center of the current FOV **300**) or toward another point along the axis of current FOV **300**. In some embodiments, the visual feedback **117** may comprise a directional indicator TU that may point in the direction of the shortest path toward center U **303** (e.g. along plane P1).

(78) Referring to FIG. 3B, instrument **120-5** is shown at location T **305** in current FOV **300**. Location T **305** is within first imaging volume **308** (shown by dashed lines in FIG. 3B). FIG. 3B also shows example visual feedback **117** in the form directional indicator **320-1**, where the length

L1 327 of the directional indicator is proportional to the distance D1 325 between location T 305 and current FOV center U 303 (or a location estimated to be the center of current FOV 300). Further, directional indicator 320-1 may be oriented toward and point in the direction of TU. (79) Referring to FIG. 3A, in some embodiments, when an instrument 120 is located at point V 310, the user may receive visual feedback 117 (e.g. on visual interface 135) indicative of direction VB 312 toward center U 303 (or an estimated center of the current FOV 300) or toward another point along the axis of current FOV 300. In some embodiments, the visual feedback 117 may comprise a directional indicator VU that may point in the direction of the shortest path toward center U 303 (e.g. along plane P1).

(80) Referring to FIG. 3C, instrument 120-5 is shown at location V 310 in current FOV 300. Location V 310 is within current FOV 300 and within a threshold of a boundary of current FOV 300 (e.g. between current FOV 300 and first imaging volume 308 (shown by dashed lines in FIG. 3C)). Specifically, instrument 120-5 is within current FOV 300 and at a distance D3 333 from the nearest boundary of FOV 300.

(81) FIG. 3C also shows example visual feedback 117 in the form of directional indicator 320-2, where the length L2 337 of the directional indicator may be: (a) proportional to the distance D2 335 between location V 310 and current FOV center U 303 (or a location estimated to be the center of current FOV 300); or (b) inversely proportional to distance (e.g. D3 333) to the nearest boundary of current FOV 300. Further, directional indicator 320-2 may be oriented toward and point in the direction of VU. As shown in FIG. 3C, instrument 120-5 may be at a distance D3 333 from the nearest boundary of current FOV 300. In some embodiments, the visual feedback 117 may include progressively changing the color of directional indicator 320-2 (e.g. from green to yellow to orange to red) as the instrument is moved toward the boundary of current FOV 300 and/or other visual feedback 117 such as blinking the visual indicator. In the example above, when the instrument is outside current FOV 300 (or partially outside current FOV 300) the color of visual indicator 320-2 may be changed to red.

(82) Referring to FIG. 3A, in some embodiments, when an instrument (e.g. instrument 120-5) is within current FOV 300 and within a threshold of the boundary of current FOV 300, (in addition to visual feedback 117 or instead of visual feedback 117) the user may receive haptic feedback 141. For example, if the instrument (e.g. instrument 120-5) is moved from point V 310 in a direction toward the boundary of current FOV 300 (e.g. in a direction away from the axis of current FOV 300 or away from center U 303), the haptic feedback 141 may comprise haptic resistance (e.g. to the movement of haptic interfaces 140). For example, the haptic resistance provided may be inversely proportional to the distance D3 333 (FIG. 3C) of the instrument (e.g. instrument 120-5) from the nearest boundary of current FOV 300. Thus, a user may encounter greater haptic feedback 141 (e.g. haptic resistance to movement) as the instrument (e.g. instrument 120-5) approaches the boundary of current FOV 300.

(83) In some embodiments, when an instrument (e.g. instrument 120-5) is located at point W 315 outside of the boundary of current FOV 300, the user may receive visual feedback 117 (e.g. on visual interface 135) indicative of direction WE 317 toward current FOV 300. In some embodiments, the visual feedback 117 may comprise a directional indicator that may point in the direction of the shortest path to bring the instrument into current FOV 300. In some embodiments, the directional indication may be colored red to indicate that current location W 315 is outside FOV 300. In some embodiments, the user may be able to provide input (e.g. via user interface 130) to indicate that automatic repositioning of one or more instruments 120 into current FOV 300 is desired. In response, robotic medical system 100 may reposition instruments 120 that are currently outside the boundary of current FOV 300, to a location within current FOV 300. In some embodiments, control model 175 may be used to determine that one or more instruments 120 are outside current FOV 300. In some embodiments, input provided by sensors (e.g. electromagnetic sensors and/or other sensors) may be used, at least in part, to determine the locations of instruments

120 that are outside current FOV **300**. In some embodiments, instruments **120** deemed as, or configured to be, preferably operated with user supervision may not be moved automatically (and/or visual feedback **117** may be provided to indicate that one or more instruments **120** have not been moved or cannot be moved automatically).

(84) Referring to FIG. 3D, instrument **120-5** is shown at location W **315** in current FOV **300**. Location W **315** is outside the boundary of current FOV **300**. Specifically, instrument **120-5** is outside current FOV **300** and at a distance D5 **343** from the nearest boundary of current FOV **300**.

(85) FIG. 3D also shows example visual feedback **117** in the form of directional indicator **320-3**, where the length L3 **347** of the directional indicator **320-3** is proportional to the distance D4 **345** between location W **315** and FOV center U **303** (or a location estimated to be the center of current FOV **300**). Thus, in the examples above, in relation to the lengths L1 **327**, L2 **337**, and L3 **347** of the directional indicators $L1 < L2 < L3$ (because $D1 < D2 < D3$).

(86) Further, directional indicator **320-3** may be oriented toward and point in the direction of WU. As shown in FIG. 3D, instrument **120-5** may be at a distance D5 **343** from the nearest boundary of current FOV **300**. In some embodiments, the visual feedback **117** may include blinking a visual indicator and/or changing the color of the visual indicator **320-3** to red when the instrument is outside current FOV **300** (or partially outside current FOV **300**).

(87) Referring to FIG. 3A, in some embodiments, when an instrument is outside the boundary of current FOV **300** (in addition to visual feedback **117** or instead of visual feedback **117**) the user may receive haptic feedback **141**. For example, if the instrument is moved from point W **315** in a direction away from the boundary of current FOV **300** (e.g. in a direction away from the axis of current FOV **300** or away from center U **303**), the haptic feedback **141** may comprise haptic resistance (e.g. to the movement of haptic interfaces **140**). For example, the haptic resistance provided may be proportional to the distance D4 **345** (FIG. 3D) of the instrument (e.g. instrument **120-5**) from the nearest boundary of current FOV **300**. Thus, a user may encounter greater haptic feedback **141** (e.g. haptic resistance to movement) as the instrument (e.g. instrument **120-5**) is moved further away from the boundary of current FOV **300**.

(88) In some embodiments, the haptic resistance to movement further away from the boundary of current FOV **300** may be greater when instrument **120-5** is outside the boundary, relative to haptic resistance to movement toward the boundary when instrument **120-5** is within the boundary of current FOV **300**. In some embodiments, the haptic resistance and/or visual indicators may be adjusted dynamically. For example, as the user moves toward the center of current FOV **300**, the haptic resistance may decrease dynamically based on the distance from the current location of instrument **120-5** to the center of current FOV **300**. Similarly, the length and color of the directional indicators may change as instrument **120-5** is moved closer to the center of current FOV **300**. For example, the length of the directional indicator may be decreased and the color changed from red to amber to green based on its current location. Conversely, as instrument **120-5** is moved in a direction away from the center of current FOV **300**, the length of the directional indicators may be dynamically increased and/or haptic resistance to movement of instrument **120-5** may be dynamically increased based on a current location of instrument **120-5** relative to FOV **300**. Further, the color of the directional indicator may be changed from green to amber to red as instrument **120-5** moves from a location within imaging volume **308** (green) to a location between imaging volume **308** and FOV **300** (amber), to a location outside FOV **300** (red).

(89) FIGS. 4A and 4B shows an exemplary flowchart related to a method **400** of facilitating user feedback based on the locations of instruments (e.g. instruments **120**) relative to a current FOV **300** of at least one image sensor (e.g. image sensors **110**) coupled to a robotic medical device (e.g. robotic medical device **200**). In some embodiments, some or all of method **400** may be implemented by one or more of: robotic medical system **100**, user interface **130**, robotic medical device control system **160**, and/or processor(s) **150**.

(90) In some embodiments, user interface **130** may comprise processor(s) **150**, which may be

coupled to robotic medical device **200**. Although shown separately in FIG. 1A, in some embodiments, robotic medical device control system **160** may comprise user interface **130** and processor(s) **150**, which may be coupled to robotic medical device **200**. The coupling of processor(s) **150** to robotic medical device **200** may be communicative (e.g. using wired or wireless communication to exchange information and commands), and/or electrical and/or or electronic, and/or electromagnetic, and/or electro-mechanical, and/or some combination of the above. For example, processor(s) **150** may interact electronically with image sensors **110** to control the capture of images and/or receive captured images. In some embodiments, processor(s) **150** and/or robotic medical device control system **160** may use one or more of electrical, electronic, electromagnetic, and/or electro-mechanical couplings to monitor and obtain sensor input, and/or to initiate the driving robotic medical device **200** and/or to control actuators that articulate and move main sheath **235**, sub-arms, image sensors **110**, instruments **120**, and/or haptic interface **140**.

(91) In some embodiments, processor(s) **150** may be distributed between robotic medical device control system **160** and user interface **130**. For example, user interface **130** may be located remotely from robotic medical device control system **160** and/or robotic medical device **200**, and exchange information (e.g. one or more of: captured images **115**, and/or sensor/actuator information **167**, and/or instrument state **165**) and commands with robotic medical device **200** and/or robotic medical device control system **160** and some or all of method **400** may be performed by user interface **130** based on the received information. In situations where there is an increased danger of risk to medical practitioners or patients (e.g. from infectious/contagious diseases etc.) user interface **130** communicatively coupled (e.g. via wired communications interface **107** and/or wireless communication interface **105**) to robotic medical device **200** that forms part of robotic medical system **100** may be used at some specified distance or through barriers (e.g. such as optically transparent but biologically protective barriers) to perform medical procedures on the patient while maintaining safety and distancing protocols.

(92) Further, in situations when skilled practitioners are unavailable (e.g. in remote locations), user interface **130** communicatively coupled (e.g. via wired communications interface **107** and/or wireless communication interface **105**) to robotic medical device **200** that forms part of robotic medical system **100** may be used telesurgically (e.g. via wired communications interface **107** or wireless communication interfaces **105**) to perform or guide performance of medical procedures (e.g. using locally available resources). For example, local medical practitioner(s) may monitor and supervise patient condition and/or the procedure being performed in accordance with any local regulations and/or safety protocols. In the example above, a medical facility deploying robotic medical device **200** may be able to use a remote first medical practitioner (e.g. at a remote first medical facility) for one medical procedure using a first user interface **130-1** (e.g. telesurgically), and use a second medical practitioner (e.g. at a remote second medical facility) for another medical procedure using a second user interface **130-2** telesurgically.

(93) As another embodiment, an integrated robotic unit, which may include processor(s) **150**, UI **130**, robotic medical device control system **160**, and robotic medical device **200**, and may perform method **400**. For example, in the integrated robotic unit, information and commands may be exchanged with robotic medical device **200** using wired communication and robotic medical device **200** may be monitored, controlled, and driven using electrical, electronic, and/or electro-mechanical techniques.

(94) In block **410**, for one or more instruments **120** coupled to a robotic medical device, **200** one or more corresponding instrument positions relative to a current FOV **300** of at least one image sensor **110** coupled to the robotic medical device may be obtained. As one example, user interface **130** may receive (e.g. from processor(s) **150** and/or robotic medical device control system **160**) the corresponding instrument positions **119** of one or more instruments **120** relative to current pose and/or relative to FOV **300** of the at least one image sensor **110**.

(95) In some embodiments, referring to FIG. 4B, in block **414**, the current FOV (e.g. current FOV

300) of at least one image sensor **110** may be obtained based on a corresponding current pose associated with at least one image sensor **110**, which may be coupled to robotic medical system **100**.

(96) In some embodiments, user interface **130** and/or a processor(s) **150** and/or robotic medical device control system **160** may determine the corresponding instrument positions **119** of one or more instruments **120** relative to current pose and/or FOV **300** of the at least one image sensor **110**. For example, input from control model **175** may be used, at least in part, to determine the pose of the at least one image sensor **110** and/or the current FOV **300**. In some embodiments, the pose of the at least one image sensor **110** may be determined based, in part, on optional sensor input **404** (e.g. provided by one or more sensors coupled to robotic medical device **200**). The optional sensor input may comprise input from electromagnetic sensors and/or electro-mechanical sensors (e.g. coupled to robotic medical device **200**), which may facilitate determination of articulation and motion of main sheath **235**, image sensors **110** and image sensor sub-arm **215**, and/or instruments **120-i** and corresponding instrument sub-arms **220**. If the total number of instruments **120** coupled to robotic medical device **200** is M , then the set of instruments may be written as $I = \{120-i | 1 \leq i \leq M\}$. In some embodiments, sensor actuator information **167** may comprise sensor input **404** to method **400**.

(97) In some embodiments, the optional sensor input **404** may further include input from 3D shape sensing fiber optic sensors, fiber optic force and/or pressure sensors such as photonic crystal fiber (PCF) sensors or Fiber Bragg Grating (FBG) sensors, or make use of scattering arising from FBG sensors or inherently present, or make use of post-process produced Rayleigh scattering.

(98) In block **418**, for one or more instruments **120** coupled to the robotic medical device **200**, one or more corresponding instrument positions relative to the current FOV **300** may be determined based on: (a) one or more captured images **115**, which may be captured by the at least one image sensor **110**, or (b) a control model **175** for the robotic medical device **200**, or (c) optional sensor input **404** from one or more sensors coupled to the robotic medical device **200**; or (d) some combination of (a)-(c) above.

(99) For example, the corresponding positions of one or more instruments relative to the current FOV **300** may be determined based, at least in part, on captured images **115** (e.g. captured by the at least one image sensor **110**), when the corresponding positions of the one or more instruments **120** lie within the current FOV **300** and the one or more instruments **120** are visible in the captured images **115**. In some embodiments, input from control model **175** may be used, at least in part, to determine the corresponding positions of the one or more instruments **120** relative to the current FOV **300**. In some embodiments, the position of the instrument **120** may be determined based in part on optional sensor input **404** from one or more sensors coupled to the robotic medical system **100**, or a combination of captured images **115** and sensor input **404**.

(100) Referring to FIG. 4A, in block **420**, the generation of a corresponding feedback for a selected instrument **120-k**, $1 \leq k \leq M$, of the one or more instruments **120** in I may be initiated, wherein the corresponding feedback may be based on a position of the selected instrument **120-k** relative to the current FOV **300**. The feedback corresponding to the selected instrument **120-k** may be provided to a user (e.g. via user interface **130**). In some embodiments, the feedback corresponding to the selected instrument **120-k** may comprise one or more of: corresponding haptic feedback **141-k**; or corresponding visual feedback **117-k**; or a combination of visual feedback **117-k** and haptic feedback **141-k**. In some embodiments, haptic feedback may be provided using haptic interface **140** (FIG. 1B) and visual feedback may be provided using visual interface **135** (FIG. 1B).

(101) In some embodiments, initiating generation the corresponding haptic feedback **141-k** may comprise initiating generation of a corresponding haptic resistance to a movement-related input for the selected instrument **120-k** in response to a determination that: (a) the position of the selected instrument **120-k** is within the current FOV **300** and within a threshold of the current FOV **300** (e.g. between first imaging volume **308** and FOV **300** in FIG. 3A) and the movement-related input is

indicative of movement of the selected instrument **120-k** toward a boundary of the current FOV **300**, or (b) the position of the selected instrument **120-k** is outside the boundary (e.g. outside FOV **300**) and the movement-related input is indicative of movement of the selected instrument **120-k** further away from the boundary.

(102) In some embodiments, the corresponding haptic resistance to movement of the selected instrument **120-k** may be inversely proportional to the distance between the boundary and the position of the selected instrument **120-k** when the position of the instrument **120-k** is within the current FOV **300** and within the threshold of the boundary (e.g. between first imaging volume **308** and FOV **300** in FIG. 3A). In some embodiments, the corresponding haptic resistance to movement of the selected instrument **120-k** may be proportional to the distance between the boundary of current FOV **300** and the position of the selected instrument **120-k** when the position of the selected instrument **120-k** is outside the boundary. In some embodiments, the corresponding haptic resistance to movement of the instrument **120-k** may be lower when the position of the selected instrument **120-k** is within the boundary of the current FOV **300** relative to the corresponding haptic resistance when the position of the selected instrument **120-k** is outside the boundary.

(103) In some embodiments, initiating generation of the corresponding visual feedback **117** may comprise displaying a corresponding directional indicator **320-k** indicating a movement direction for the selected instrument **120-k** to bring the selected instrument **120-k** (a) further into the current FOV **300** in response to a determination that the position of the selected instrument **120-k** is within the current FOV **300** and within a threshold of a boundary of the current FOV **300** (e.g. between first imaging volume **308** and FOV **300** in FIG. 3A), or (b) into the current FOV **300** in response to a determination that the position of the selected instrument **120-k** is outside the boundary of the current FOV **300**. In some embodiments, a magnitude of the directional indicator (e.g. length of the corresponding directional indicator) **320-k** may be proportional to the distance between the position of the selected instrument **120-k** and a specified location (e.g. a center of current FOV **300** or a point on a central axis of FOV **300**, or another location in first imaging volume **308**) within the current FOV **300**.

(104) In some embodiments, method **400** may further comprise (e.g. after block **420** or block **430**) selectively disabling functionality associated with the selected instrument **120-k**, when the instrument **120-k** is outside the current FOV **300**. For example, the selected instrument **120-k** (e.g. cautery knife **120-4**, for $k=4$) may be deactivated and/or disabled (or configured to be deactivated or disabled) when outside the boundary of the current FOV **300** to prevent accidental use, inadvertent contact with tissue, and/or to increase safety.

(105) In some embodiments, method **400** may comprise (e.g. after block **420** or block **430**) automatically moving the selected instrument **120-k** into a position within the current FOV **300** in response to user input when the position of the selected instrument **120-k** is outside a boundary of the current FOV **300**.

(106) FIG. 5 shows an exemplary computing subsystem **500** to facilitate instrument visualization. Computing subsystem **500** may form part of robotic medical system **100**, and/or robotic medical device **200**, and/or another robotic device and facilitate instrument visualization. For example, computing subsystem **500** may be part of robotic medical system **100**. In some embodiments, computing subsystem **500** form part of user interface **130** and/or robotic medical device control system **160**, and/or and may be coupled to robotic medical device **200**. In some embodiments, computing subsystem **500** may perform some or all of methods shown in FIGS. 4A and 4B.

(107) As shown in FIG. 5, computing subsystem **500** may include processor(s) **150**, memory **170**, and communications interface **502**, which may be connected to each other using connections **506**. Connections **506** may take the form of buses, lines, fibers, electronic interfaces, links, etc., which may operationally couple the above components.

(108) Communications interface **502** may be capable of wired (e.g. using wired communications interface **107**) or wireless (e.g. using wireless communication interface **105**) communications with

another medical device (e.g. robotic medical device control system **160**, user interface **130**, and/or robotic medical device **200**). Captured images **115**, instrument state **165**, sensor/actuator information **167**, etc., may be received over communications interface **502**. User input may also be transmitted (e.g. when computing subsystem is part of user interface **130**) or received (e.g. when computing subsystem forms part of robotic medical device control system **160**) using communications interface **502**. In some embodiments, communications interface **502** may be used to receive, transmit, or relay signals related to user feedback (haptic and/or visual to appropriate components of robotic medical system **100**). Wireless communication may include communication over Wireless Personal Area Networks (WPANs), which may be based on the IEEE 802.15x standards, Wireless Local Area Networks (WLANs), which may be based on the IEEE 802.11 standards, and/or over Wireless Wide Area Networks (WWAN), which may be based on cellular communication standards such as a Fifth Generation (5G) network, or Long Term Evolution (LTE). (109) Computing subsystem **500** may also include control interface **508**, which may provide control input (e.g. activate, select, deploy, deactivate, retract, etc.) and command input to drive robotic medical device **200** (e.g. main sheath **235**, sub-arms, image sensors **110**, and/or instruments **120**). In some embodiments, control interface **508** may also output haptic feedback **141** (e.g. based on the position of instrument **120** relative to current FOV **300** of image sensors **110** and user input related to instrument movement direction). Control interface **508** may communicate with processor(s) **150** and may be controlled by processor(s) **150**.

(110) Computing subsystem **500** may also include display interface **510**, which may interact with a display (e.g. a 3D or stereoscopic display) to provide visual feedback **117** (e.g. based on the position of instrument **120** relative to current FOV **300** of image sensors **110**). For example, display interface may generate graphics, and/or other visualization, which may augment or overlay the captured images. In some embodiments, display interface **510** may be used to generate visual feedback **117** to display directional indicator **320** indicating a corresponding movement direction to bring one or more instruments **120** further into the current FOV **300**. Display interface **510** may communicate with processor(s) **150** and may be controlled by processor(s) **150**.

(111) In some embodiments, memory **170** may comprise main or primary memory (e.g. RAM) and storage **560**. Program code may be stored in memory **170**, and read and executed by processor(s) **150** to perform the techniques disclosed herein. Storage **560** may include ROM, EPROM, NVRAM, flash memory, secondary storage, and other computer readable media (e.g. fixed and/or removable drives, optical disks, etc.). Computer-readable media may be encoded with databases, data structures, etc. and/or with computer programs. By way of example, and not limitation, such computer-readable media may also include CD-ROM, memory cards, portable drives, or other optical disk storage, magnetic disk storage, solid state drives, other storage devices, or any other medium that can be used to store desired program code in the form of instructions and/or data structures and that can be accessed by a computer.

(112) In some embodiments, images captured by image sensors **110**, instrument state **165**, sensor/actuator information **167**, motion control/instrument control input **162**, and other information pertaining to robotic medical device **200** may be stored in memory **170** for operational, training, logging, and other purposes. For example, based on user input and configuration information **177**, a procedure performed by robotic medical device **200** may be recorded and replayed/analyzed at a subsequent time. As another example, the procedure may be live streamed via communications interface **502** (e.g. to local and/or remote medical practitioners during telesurgery, and/or for educational or training purposes).

(113) Memory **170** may include control model **175**, which may be used by processor(s) **150** to determine the positions of instruments **120** relative to image sensors **110**. In some embodiments, control model **175** may include calibrated instrument control models, which may be used to estimate an instrument position (e.g. relative to image sensors **110**) based on one or more of: motion control/instrument control input **162**, configuration information **177**, and/or motion-related

user input **143**. In some embodiments, control model **175** may also use information from one or more sensors (when present) in robotic medical device **200**.

(114) Memory **170** may include configuration information **177**, which may provide information pertaining to the instruments on robotic medical device **200**, image sensor configuration (e.g. lens focal length and other parameters), user preferences (e.g. sensitivity to user movement, the desired level of haptic feedback **141**, automatic maintenance of FOV over instruments **120**, indicating whether functionality of one or more instruments **120** is to be disabled when not in the FOV of image sensors **110**, display parameters, etc.) and/or an operational configuration or mode of operation of robotic medical system **100**.

(115) The methodologies described herein may be implemented in hardware, firmware, software, or any combination thereof. For a hardware implementation, the processor(s) **150** may be implemented within one or more application specific integrated circuits (ASICs), digital signal processors (DSPs), image processors, digital signal processing devices (DSPDs), programmable logic devices (PLDs), field programmable gate arrays (FPGAs), processors, controllers, micro-controllers, microprocessors, electronic devices, other electronic units designed to perform the functions described herein, or any combination thereof. In some embodiments, processor(s) **150** may include capabilities to: determine a current pose of image sensor **110** based on sensor input, determine a FOV associated with image sensors **110** determine a FOV **300** for image sensors **110** based on a current pose, process images, determine and track corresponding positions of instruments **120** relative to a current image sensor FOV **300**, provide input to control haptic feedback **141**, provide visual feedback **117**, and provide input to control actuators on robotic medical device **200**. Processor(s) **150** may also include functionality perform other well-known computer vision and image processing functions such as feature extraction from images, image comparison, image matching, object recognition and tracking, image compression and decompression, etc.

(116) For example, captured images **115** may be processed using object recognition and/or tracking techniques by processor(s) **150** to determine a location of one or more instruments **120** in the current FOV **300** of image sensor **110**. In some embodiments, based on configuration information **177** (e.g. as set or invoked by a user), processor(s) **150** may use object tracking techniques to keep an instrument in the current FOV **300** of image sensors **110** during a procedure (or portion of a procedure). For example, the location of instrument **120** may be tracked over a sequence of captured images **115**. Object tracking techniques may use one or more of: control model **175** (which may receive input from sensors coupled to or associated instruments **120**), the known instrument form factor (e.g. grasper, cauterizing knife, etc.), markers on instruments to facilitate object recognition, identification, and tracking, and/or other instrument characteristics. Object tracking techniques may be used in conjunction with image processing and computer vision techniques and sensor input to locate an instrument in an image. Processor(s) **150** may then initiate the movement of image sensors **110** to keep the instrument **120** in the FOV of image sensors **110**. When instruments are visible, because the form factors of instruments coupled to robotic medical device **200** are known, object recognition and tracking techniques may be applied to identify visible instruments in a sequence of image(s) captured by image sensors **110** and determine their respective locations relative to the image sensors and/or the corresponding image sensor FOV. When instruments **120** are not currently visible, sensor input along with prior known instrument positions may be used by control model **175** to determine instrument positions at a current time.

(117) Although the present disclosure is described in connection with specific embodiments for instructional purposes, the disclosure is not limited thereto. Various adaptations and modifications may be made to the disclosure without departing from the scope. Therefore, the spirit and scope of the appended claims should not be limited to the foregoing description.

Claims

1. A processor-implemented method comprising: (a) obtaining, for one or more instruments coupled to a robotic medical device, one or more corresponding instrument positions and instrument movement directions relative to a current field of view (FOV) of at least one image sensor coupled to the robotic medical device; and (b) initiating, for a selected instrument of the one or more instruments, generation of a corresponding feedback to a user, wherein the corresponding feedback is in real-time and is based on an obtained position of the selected instrument relative to the current FOV, wherein the corresponding feedback includes a directional indicator indicating the corresponding instrument movement directions, wherein the corresponding feedback comprises a corresponding visual feedback, wherein initiating generation of the corresponding visual feedback comprises: initiating display of the directional indicator indicating a movement direction for the selected instrument to bring the selected instrument: (i) further into the current FOV in response to a determination that the obtained position of the selected instrument is within the current FOV and within a threshold of a boundary of the current FOV, or (ii) into the current FOV in response to a determination that the obtained position of the selected instrument is outside the boundary of the current FOV.
2. The method of claim 1, wherein obtaining the one or more corresponding instrument positions relative to the current FOV comprises: (i) determining, based on a corresponding current pose associated with the at least one image sensor, the current FOV of the at least one image sensor; and (ii) determining, for the one or more instruments, the one or more corresponding instrument positions relative to the current FOV based on: (A) one or more images captured by the at least one image sensor, (B) a control model for the robotic medical device, (C) sensor input from one or more sensors coupled to the robotic medical device, or (D) a combination thereof.
3. The method of claim 2, wherein the one or more sensors comprise: (i) electromagnetic sensors, (ii) electro-mechanical sensors, wherein the electro-mechanical sensors comprise micro electro-mechanical sensors (MEMS), (iii) 3D shape sensing fiber optic sensors, (iv) fiber optic force sensors, (v) fiber optic pressure sensors, wherein the fiber optic force sensors and the fiber optic pressure sensors comprise one or more of: photonic crystal fiber (PCF) sensors or Fiber Bragg Grating (FBG) sensors, or (vi) a combination thereof.
4. The method of claim 1, wherein the corresponding feedback further comprises a corresponding haptic feedback.
5. The method of claim 4, wherein initiating generation of the corresponding haptic feedback comprises: initiating generation of a corresponding haptic resistance to a corresponding movement-related user input for a movement of the selected instrument in response to a determination that: (i) the obtained position of the selected instrument is within the current FOV and within a threshold of the current FOV and the corresponding movement-related user input indicates the movement of the selected instrument is toward a boundary of the current FOV, or (ii) the obtained position of the selected instrument is outside the boundary and the corresponding movement-related user input indicates the movement of the selected instrument is further away from the boundary.
6. The method of claim 5, wherein: the corresponding haptic resistance is inversely proportional to a distance between the boundary and the obtained position of the selected instrument when the obtained position of the selected instrument is within the current FOV and within the threshold of the boundary; and the corresponding haptic resistance is proportional to the distance between the boundary and the obtained position of the selected instrument when the obtained position of the selected instrument is outside the boundary.
7. The method of claim 6, wherein the corresponding haptic resistance is lower when the obtained position of the selected instrument is within the boundary of the current FOV relative to the corresponding haptic resistance when the obtained position of the selected instrument is outside the

boundary.

8. The method of claim 1, wherein initiating display of the directional indicator is to bring the selected instrument further into the current FOV in response to the determination that the obtained position of the selected instrument is within the current FOV and within the threshold of a boundary of the current FOV.

9. The method of claim 1, wherein a magnitude of the directional indicator is: proportional to a distance between the obtained position of the selected instrument and a specified location within the current FOV.

10. The method of claim 1, further comprising: automatically disabling a functionality associated with the selected instrument, when the selected instrument is outside a boundary of the current FOV.

11. The method of claim 1, further comprising: initiating automatic movement of the selected instrument into a position within the current FOV, in response to a user input, when the obtained position of the selected instrument is outside a boundary of the current FOV.

12. A robotic medical device comprising: (a) at least one image sensor; (b) one or more instruments; and (c) a processor communicatively coupled to the at least one image sensor and the one or more instruments, wherein the processor is configured to: (i) obtain, for the one or more instruments, one or more corresponding instrument positions and instrument movement directions that are based in part on a current image of the one or more instruments captured by the at least one image sensor, the at least one image sensor having a current field of view (FOV), and (ii) initiate, for a selected instrument of the one or more instruments, generation of a corresponding feedback to a user, wherein the corresponding feedback is based on a prior position of the selected instrument relative to the current FOV, wherein the corresponding feedback includes a directional indicator indicating the corresponding instrument movement directions, wherein the corresponding feedback comprises a corresponding visual feedback, wherein to initiate generation of the corresponding visual feedback, the processor is configured to: initiate display of a corresponding directional indicator indicating a movement direction for the selected instrument to bring the selected instrument: (i) further into the current FOV in response to a determination that the prior position of the selected instrument is within the current FOV and within a threshold of a boundary of the current FOV, or (ii) into the current FOV in response to a determination that the prior position of the selected instrument is outside the boundary of the current FOV.

13. The robotic medical device of claim 12, further comprising: one or more sensors coupled to the processor, wherein to obtain the one or more corresponding instrument positions based in part on the current FOV of the at least one image sensor, the processor is further configured to: (i) determine, based on a corresponding current pose associated with the at least one image sensor, a current FOV of the at least one image sensor, and (ii) determine the one or more corresponding instrument positions relative to the current FOV based on: (A) one or more images captured by the at least one image sensor, (B) a control model for the robotic medical device, (C) sensor input from the one or more sensors, or (D) a combination thereof.

14. The robotic medical device of claim 13, wherein the one or more sensors comprise: (i) electromagnetic sensors, (ii) electro-mechanical sensors, wherein the electro-mechanical sensors comprise micro electro-mechanical sensors (MEMS), (iii) 3D shape sensing fiber optic sensors, (iv) fiber optic force sensors, (v) fiber optic pressure sensors, wherein the fiber optic force sensors and the fiber optic pressure sensors comprise one or more of: photonic crystal fiber (PCF) sensors or Fiber Bragg Grating (FBG) sensors, or (vi) a combination thereof.

15. The robotic medical device of claim 12, wherein the corresponding feedback further comprises a corresponding haptic feedback.

16. The robotic medical device of claim 15, wherein to initiate generation of the corresponding haptic feedback, the processor is configured to: initiate generation of a corresponding haptic resistance to a corresponding movement-related user input for a movement of the selected

instrument in response to a determination that: (i) the prior position of the selected instrument is within the current FOV and within a threshold of the current FOV and the movement-related user input indicates that the selected instrument is to be moved toward a boundary of the current FOV, or (ii) the prior position of the selected instrument is outside the boundary and the movement-related user input indicates that the selected instrument is to be moved further away from the boundary.

17. A processor-implemented method comprising: (a) obtaining, for one or more instruments coupled to a robotic medical device, one or more corresponding instrument positions and instrument movement directions relative to a current field of view (FOV) of at least one image sensor coupled to the robotic medical device; and (b) initiating, for a selected instrument of the one or more instruments, generation of a corresponding feedback to a user, wherein the corresponding feedback is in real-time and is based on an obtained position of the selected instrument relative to the current FOV, wherein the corresponding feedback includes a directional indicator indicating the corresponding instrument movement directions, wherein the corresponding feedback comprises a corresponding haptic feedback, wherein initiating generation of the corresponding haptic feedback comprises: initiating generation of a corresponding haptic resistance to a corresponding movement-related user input for a movement of the selected instrument in response to a determination that: (i) the obtained position of the selected instrument is within the current FOV and within a threshold of the current FOV and the corresponding movement-related user input indicates the movement of the selected instrument is toward a boundary of the current FOV, or (ii) the obtained position of the selected instrument is outside the boundary and the corresponding movement-related user input indicates the movement of the selected instrument is further away from the boundary.

18. The method of claim 17, wherein initiating generation of the corresponding haptic resistance is in response to a determination that: the obtained position of the selected instrument is within the current FOV and within a threshold of the current FOV and the corresponding movement-related user input indicates the movement of the selected instrument is toward a boundary of the current FOV.

19. The method of claim 17, wherein initiating generation of the corresponding haptic resistance is in response to a determination that: the obtained position of the selected instrument is outside the boundary and the corresponding movement-related user input indicates the movement of the selected instrument is further away from the boundary.
